Synthetic MRI Generation with Latent Diffusion Models

Gian Favero

Gian Favero

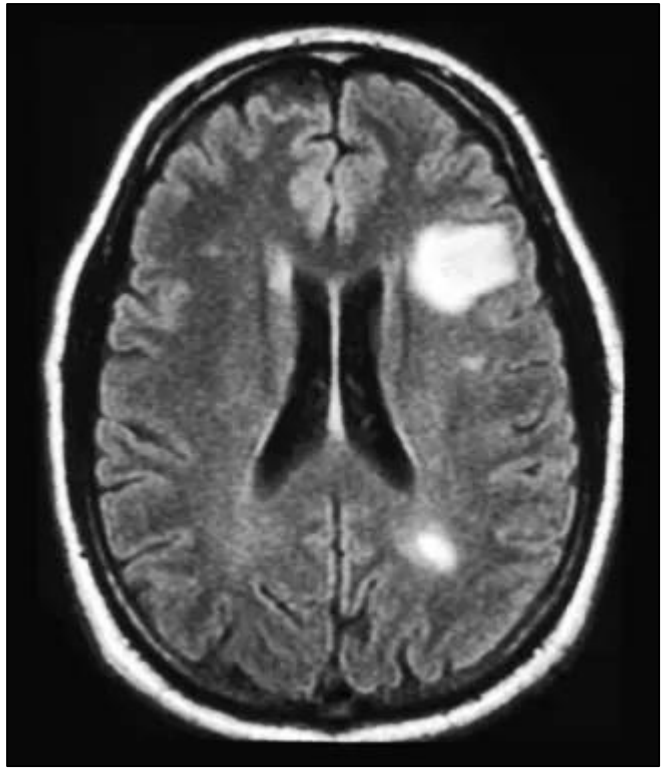
November 28th, 2023

Table of Contents

1. Introduction/Motivation
2. Problem Definition
3. Methodology
4. Results/Evaluations
5. Conclusions

Introduction/Motivation

Background: Multiple Sclerosis

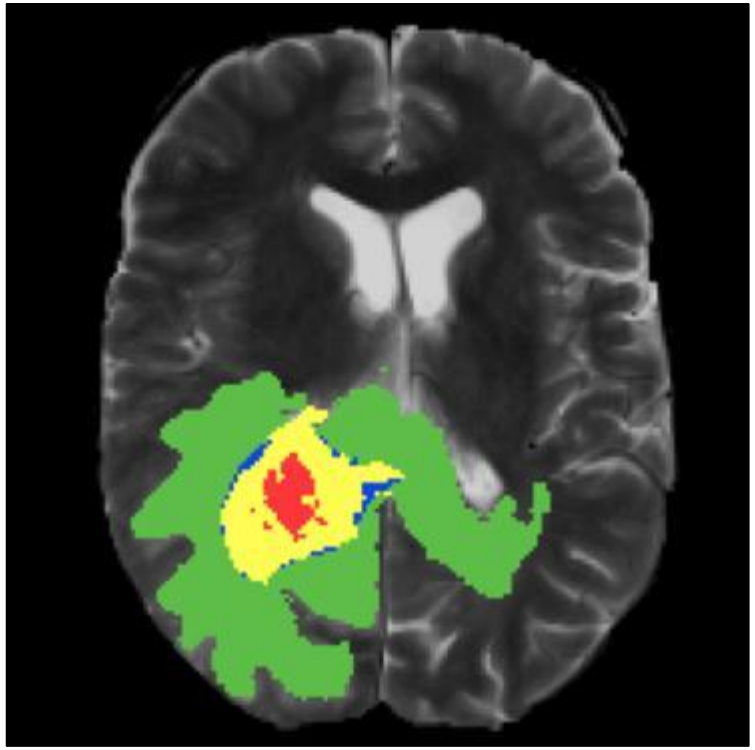


https://img.medscapestatic.com/pi/meds/ckb/81/29681tn.jpg

- Multiple sclerosis (MS) is a chronic neurological disease
- Identified by appearance of lesions in MRI scans of the brain
- Treatments available, **no cure**

Background: Image-Based Medicine

- Image-based medicine is an evolving field in modern science
- Computer vision solutions (DCNNs) emerging across medical landscape
- Applications:
  - Pathology detection
  - Projection of disease trajectory
  - Individual treatment effect estimation



https://paperswithcode.com/task/brain-tumor-segmentation

Background: Probabilistic Vision Group

- PVG (McGill) applies computer vision and ML frameworks in neurology
- Development of DL models
- Focus in multiple sclerosis (MS)

PVG

Probabilistic Vision Group

https://www.cim.mcgill.ca/~pvg/

Motivation: Data Availability

- Access to data is paramount in AI
- Lack of high-quality datasets creates hurdle
- Hurdle is exaggerated in medical contexts
  - Difficulty recruiting patients
  - High cost of acquisition
  - Privacy concerns
  - Bias

https://paperswithcode.com/dataset/imagenet

Problem Definition

A lack of large, high-quality, bias-free, and public MRI datasets exist for multiple sclerosis.

Potential Solution: Synthetic Data

Alleviate barrier to data with synthetic image generation

Small Sample Size

- Use synthetic data as an augmentation
- Increase size of existing datasets

Bias

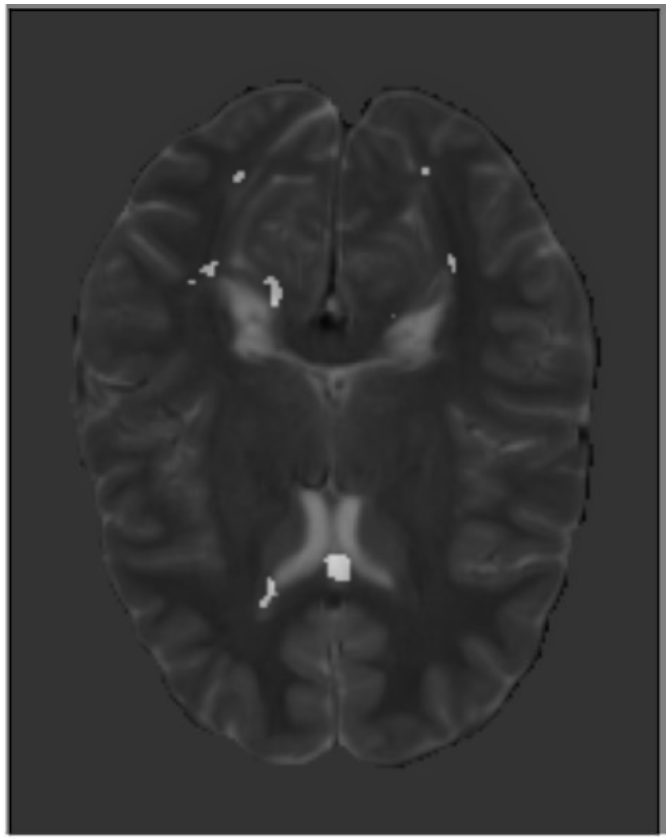
- Create debiasing synthetic data
- Fill gaps in dataset

Privacy

- Create privacy-preserving data
- Provide access for increased research

Constraints

- ☑ Anatomical plausibility (Qualitative, Classification)
- ☑ Uniqueness (SSIM)
- ☑ Pathology present (Downstream)



Methodology

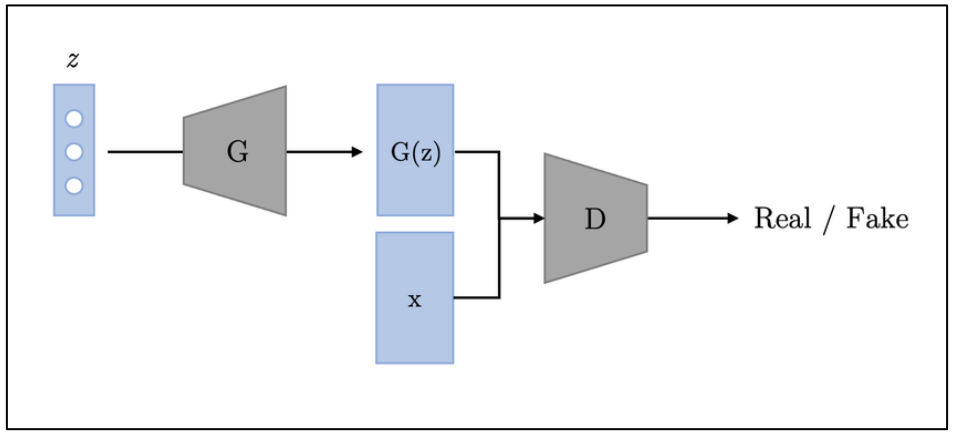
McGill

Related Work: GANs

Generative Adversarial Nets: GAN [1], StyleGAN [2]

Advantages

- Realistic generation
- High-resolution



https://www.researchgate.net/figure/A-simple-illustration-of-a-GAN-architecture

Disadvantages

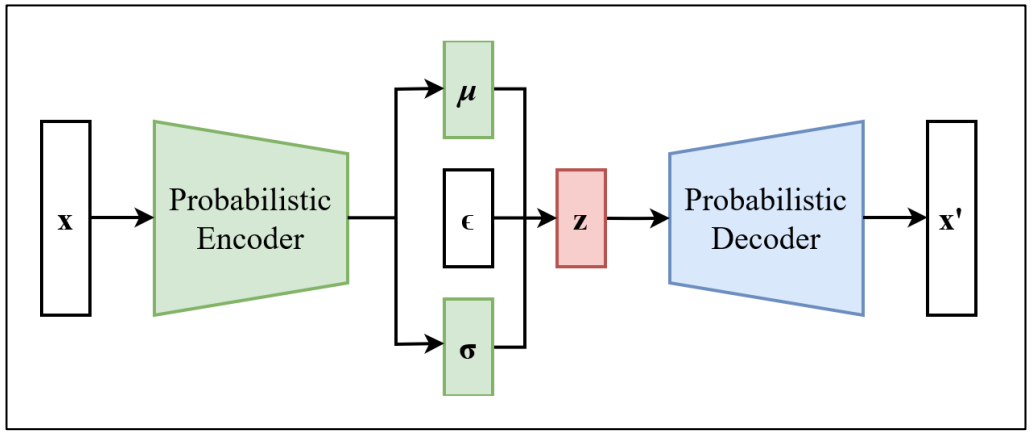
- Mode collapse
- Training instability

Related Work: VAEs

Variational Autoencoders: VAE [3], VQ-VAE [4]

Advantages

- Distribution learned*
- Smooth interpolation

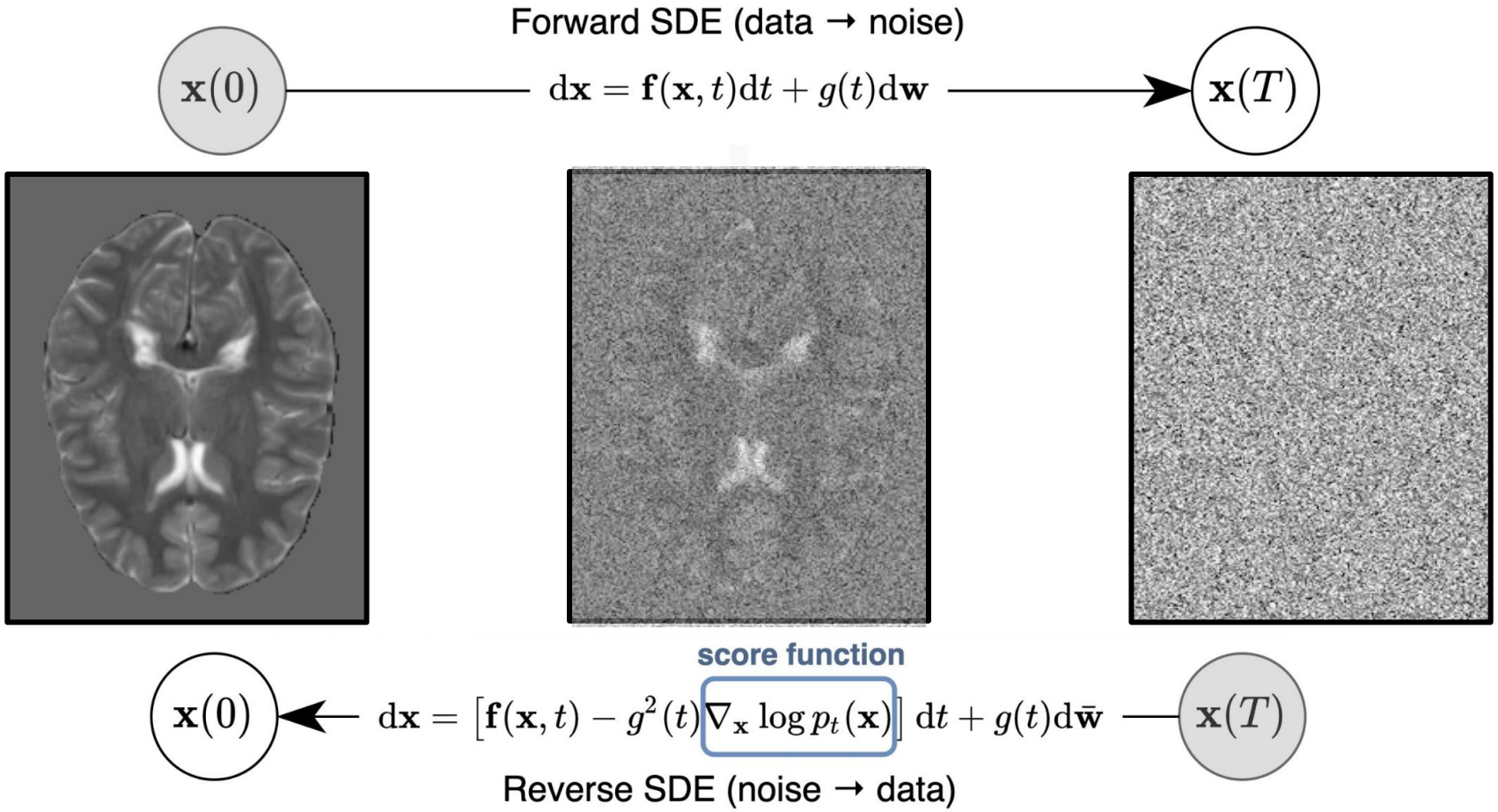


https://medium.com/mlearning-ai/variational-auto-encoders

Disadvantages

- Blurry generation

Related Work: DPMs

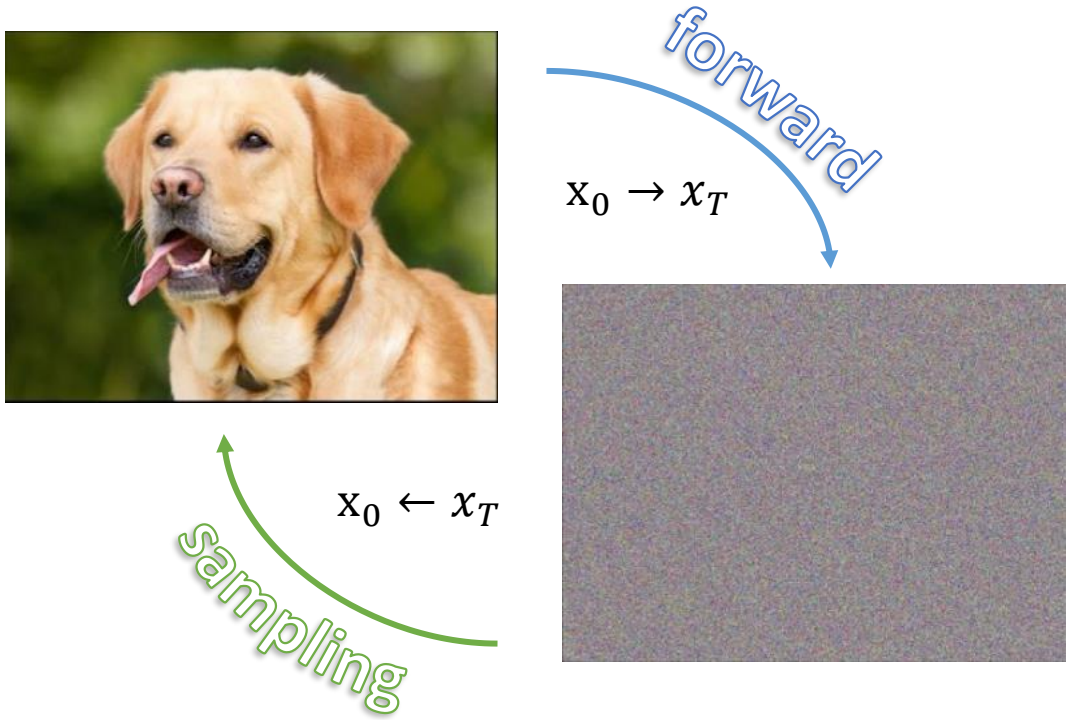


Related Work: DPMs

Diffusion Probabilistic Models: DDPM [5], DDIM [6]

Advantages

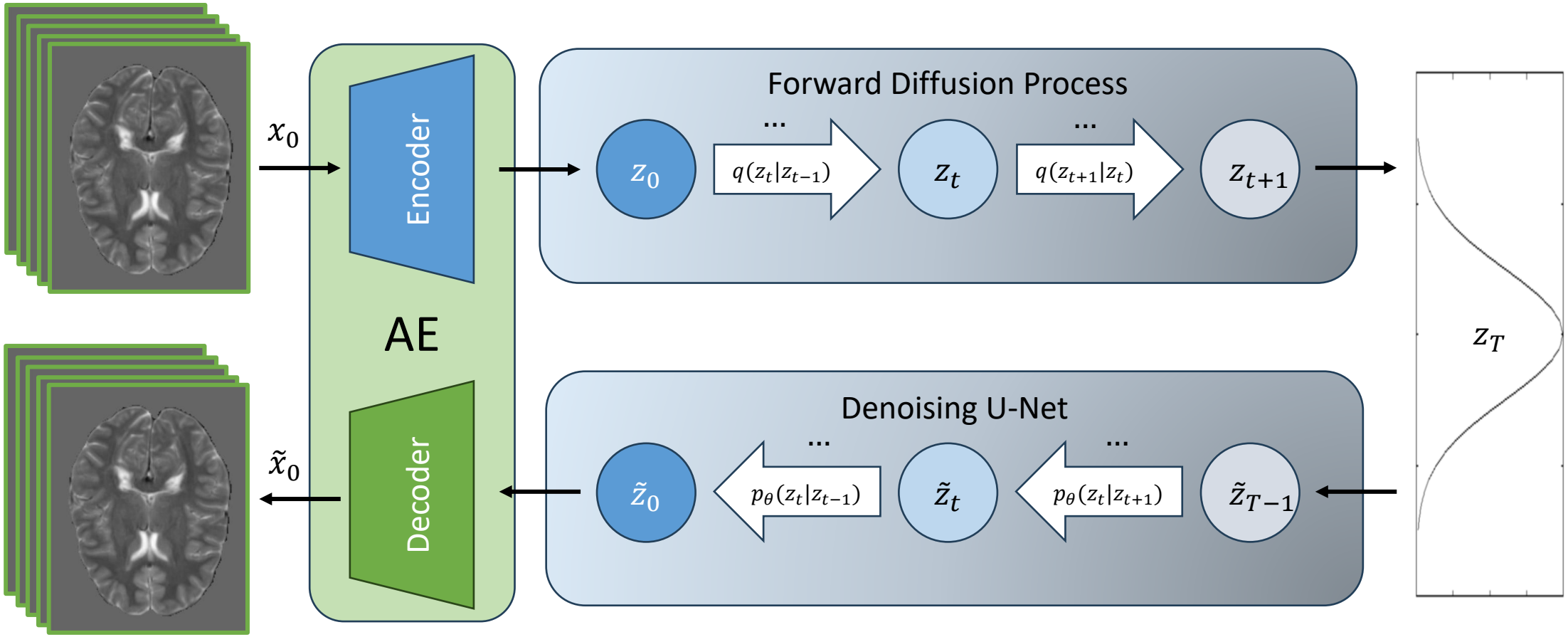
- Stable training
- Higher fidelity output



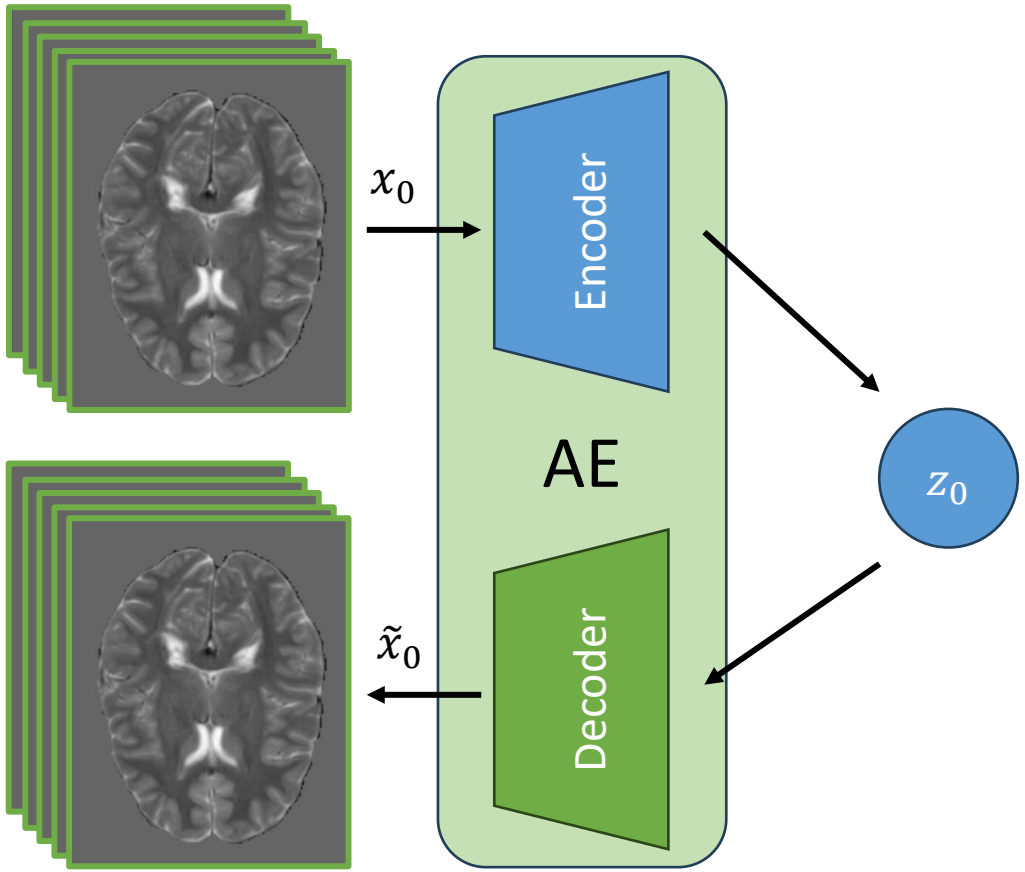
Disadvantages

- Computation-hungry
- Long sampling time

Latent Diffusion Model [6]

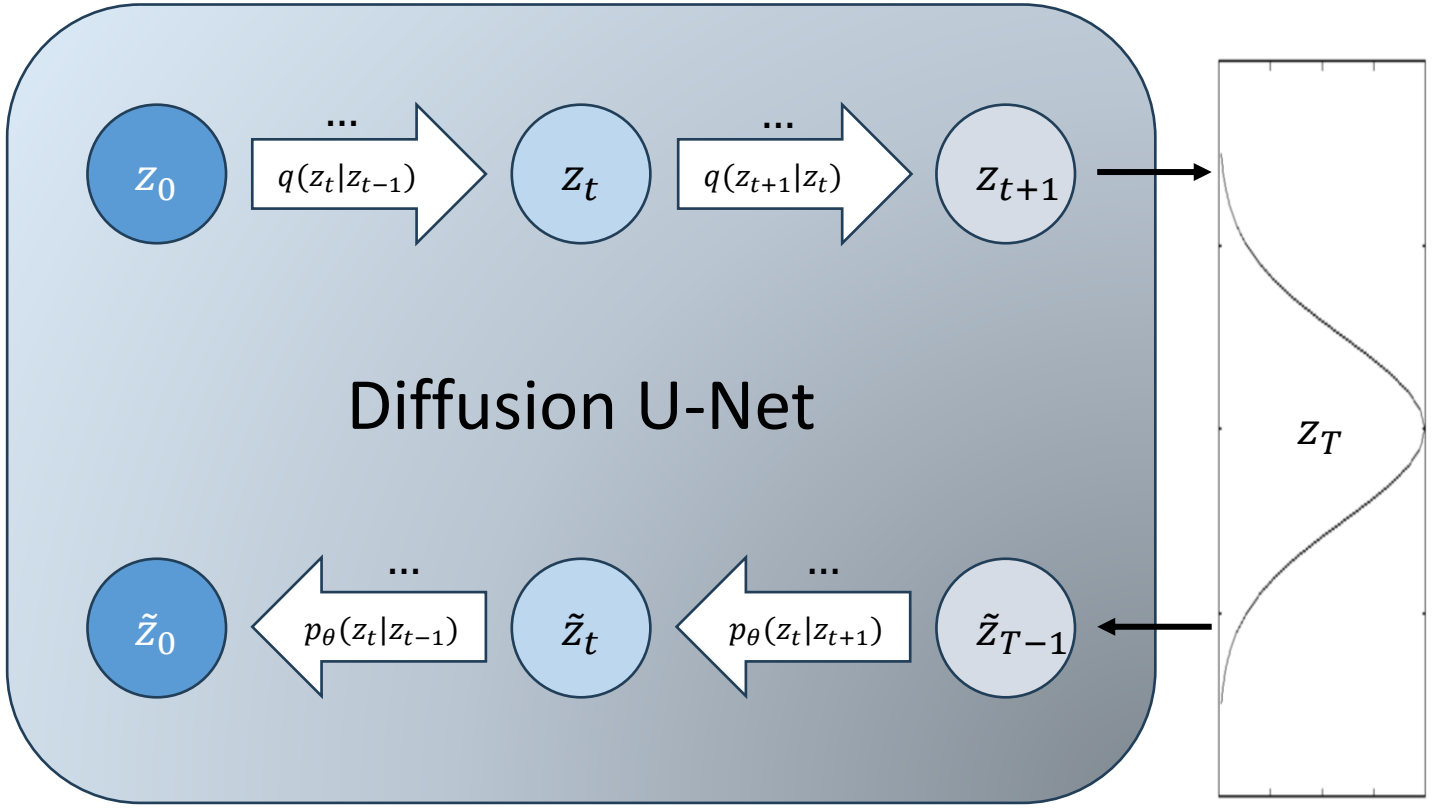


Latent Diffusion Model



- Independently trained autoencoder model reduces image to latent
- Operating in latent space reduces computational burden
- Modularity:
  - Autoencoder with KL-reg
  - VQ-VAE

Latent **Diffusion Model**

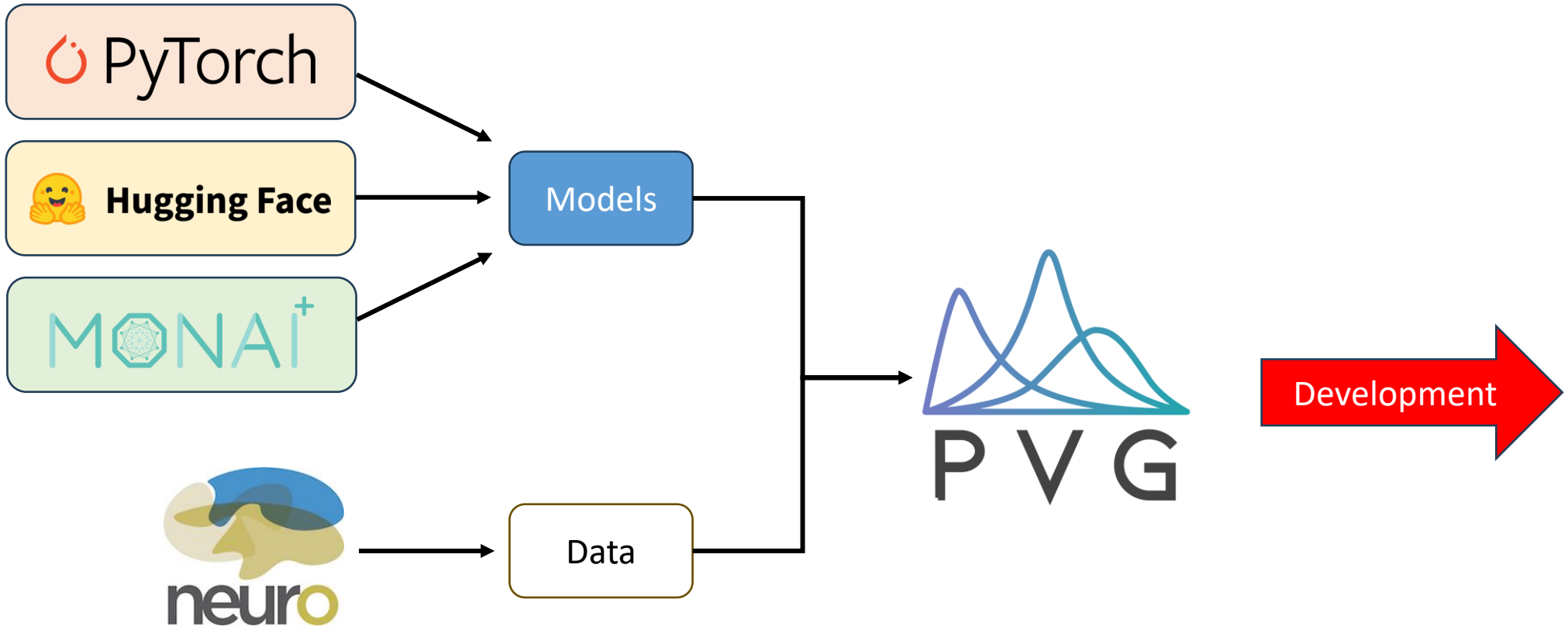


- U-Net used to learn noise added at each timestep, t
- Forward and denoising process occur in z space

Implementation

McGill

Implementation



Models

PyTorch

Hugging Face

MONAI+

- Open-source frameworks leveraged for implementation
- Modular aspect of LDM allows for mixing and matching of models

Dataset

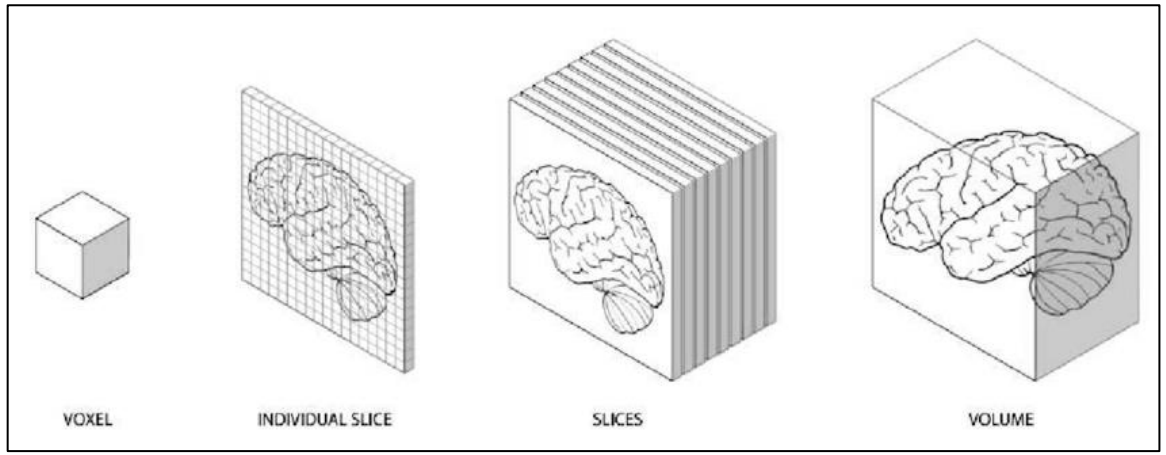
- **I**nternational **P**rogressive **M**ultiple **S**clerosis **A**lliance (IPMSA) grouped together data from several drug companies
- 3D MRI data collected from MS drug trials (2005-2020)
- Patient MRI images consist of up to 5 sequences, lesion maps
- Subset of **4500** samples from unique patients used

Simplifications

2D slice of 3D MRI

T2-weighted sequence

No bias to correct



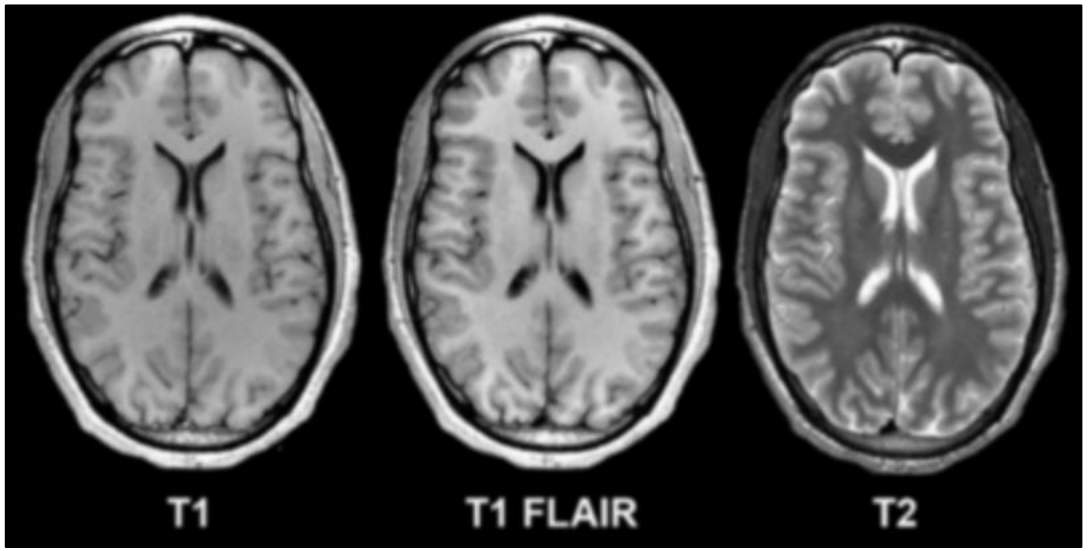
https://www.researchgate.net/figure/MRI-Voxels

Simplifications

2D slice of 3D MRI

T2-weighted sequence

No bias to correct



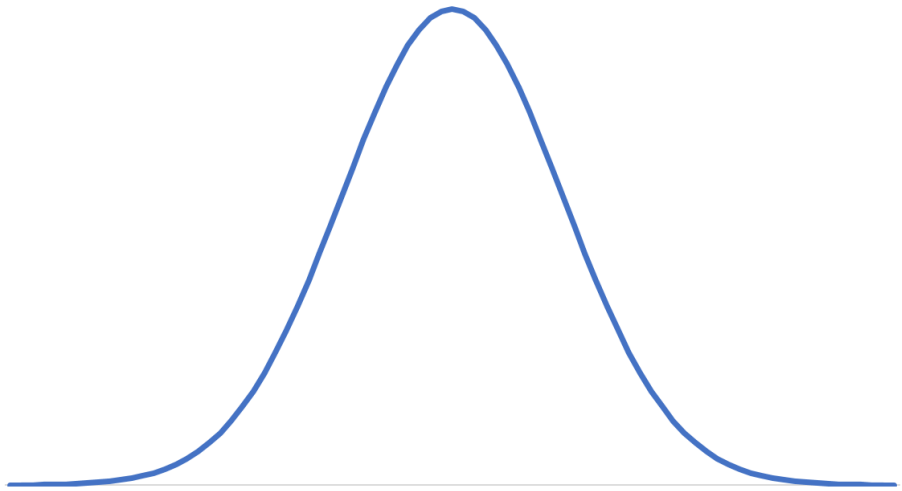
https://www.researchgate.net/publication/synthetic-MRI-in-which-a-standardized-set-of.png

Simplifications

2D slice of 3D MRI

T2-weighted sequence

No bias to correct

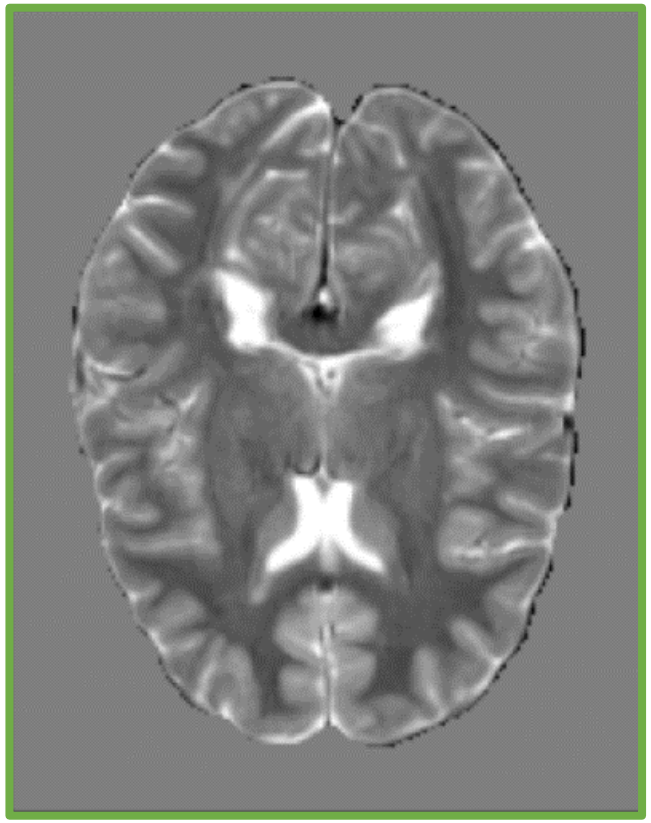


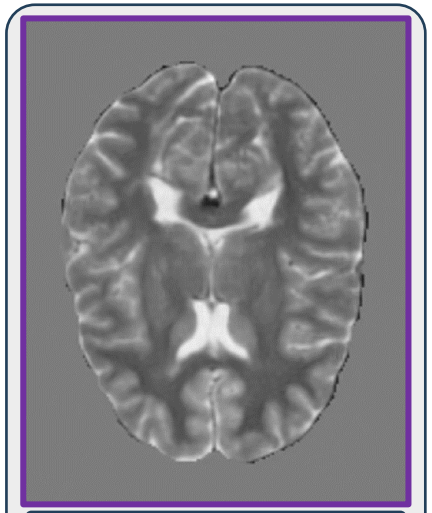
https://hammychang.medium.com/normal-distributions-2e395d596c42

Results/Evaluations

McGill

Autoencoder: Results

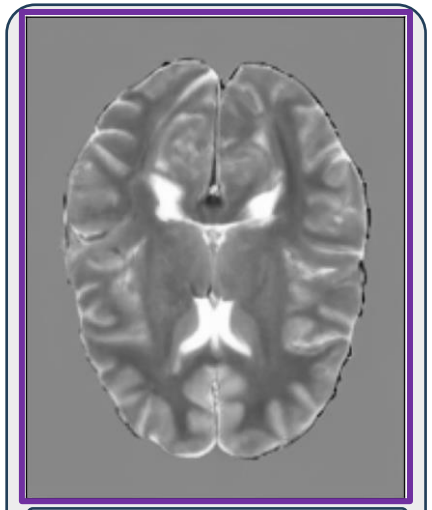




AutoencoderKL

MONAI+

Trained



SDXL-VAE

🤗 **Hugging Face**

Pre-Trained

Autoencoder: Quantitative Metrics

- **S**tructural **S**imilarity **I**ndex **M**easure [8]
  - Perceptual metric that quantifies the similarity between two images
  - Measured between 0 and 1
  - **SSIMMetric** MONAI

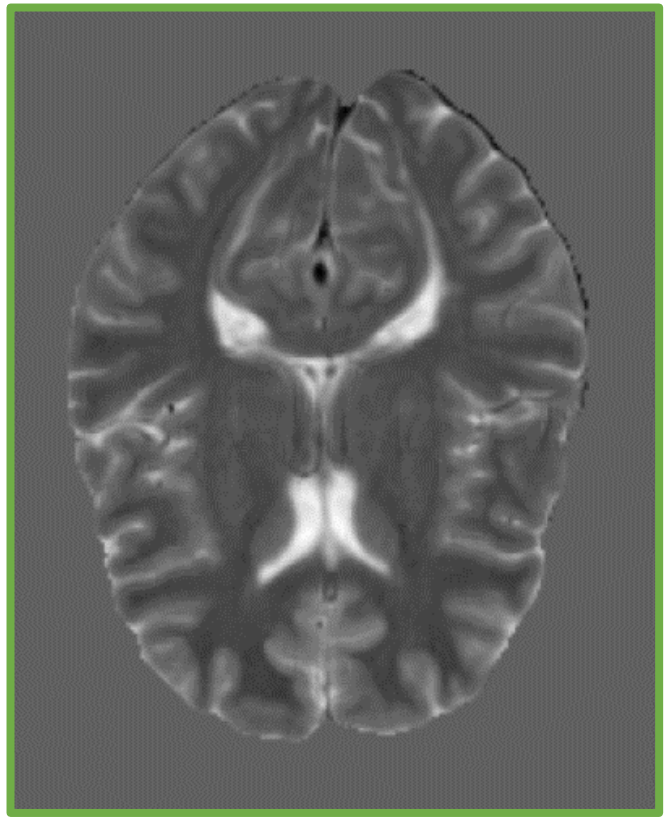
- Reconstruction Loss
  - Mean absolute error between two images
  - F.**L1_Loss** PyTorch

Autoencoder: Quantitative Metrics

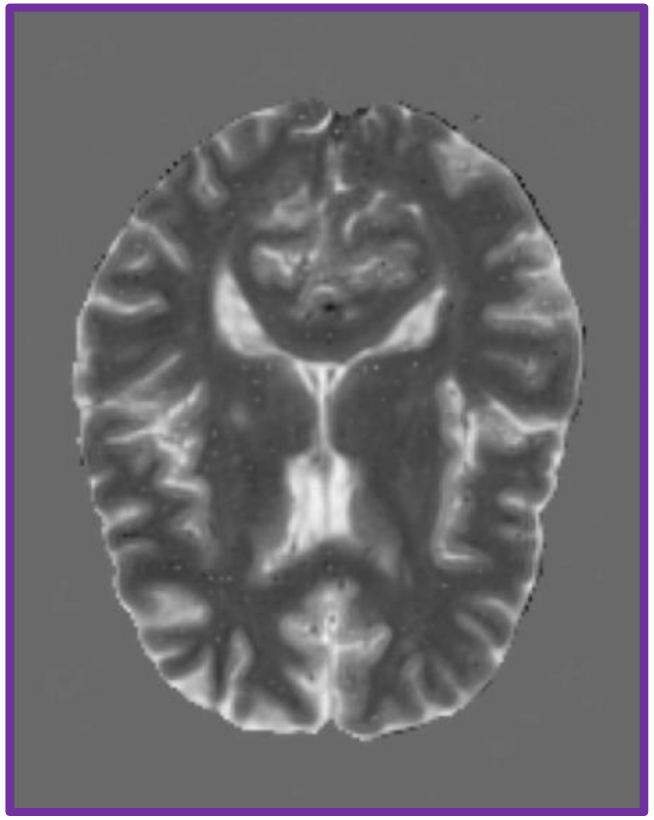
Table 1: Performance Evaluation of Autoencoders

Model	Latent Dim.	SSIM	Reconstruction Loss
AutoencoderKL	3	**0.9866**	**0.0091**
SDXL-VAE	4	0.9599	0.0126

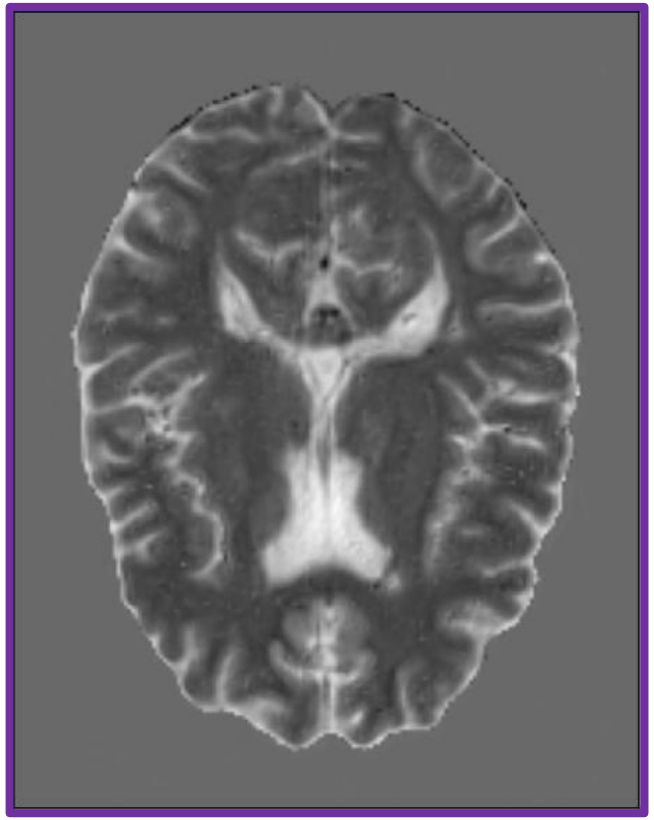
LDM: Results



Real



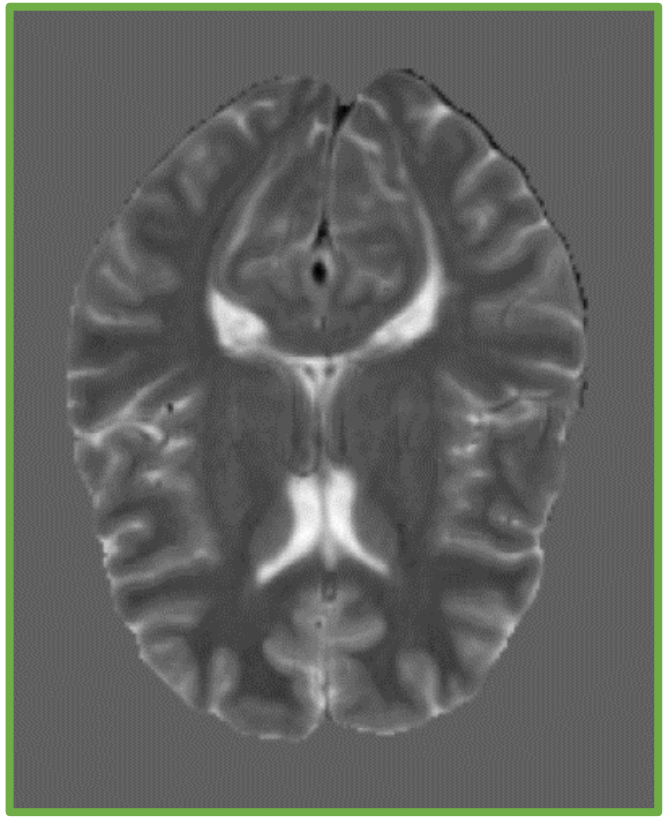
Generated



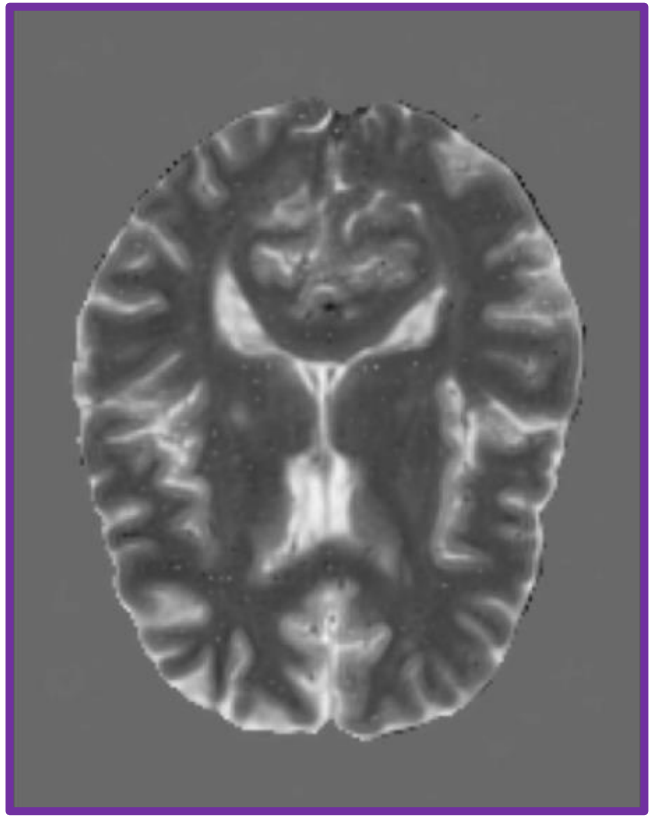
Generated

LDM: Anatomical Plausibility

- Visually appear to be reasonable
- Quantitative metrics:
  - FID vs baseline
  - Classifier
- Further exploration needed



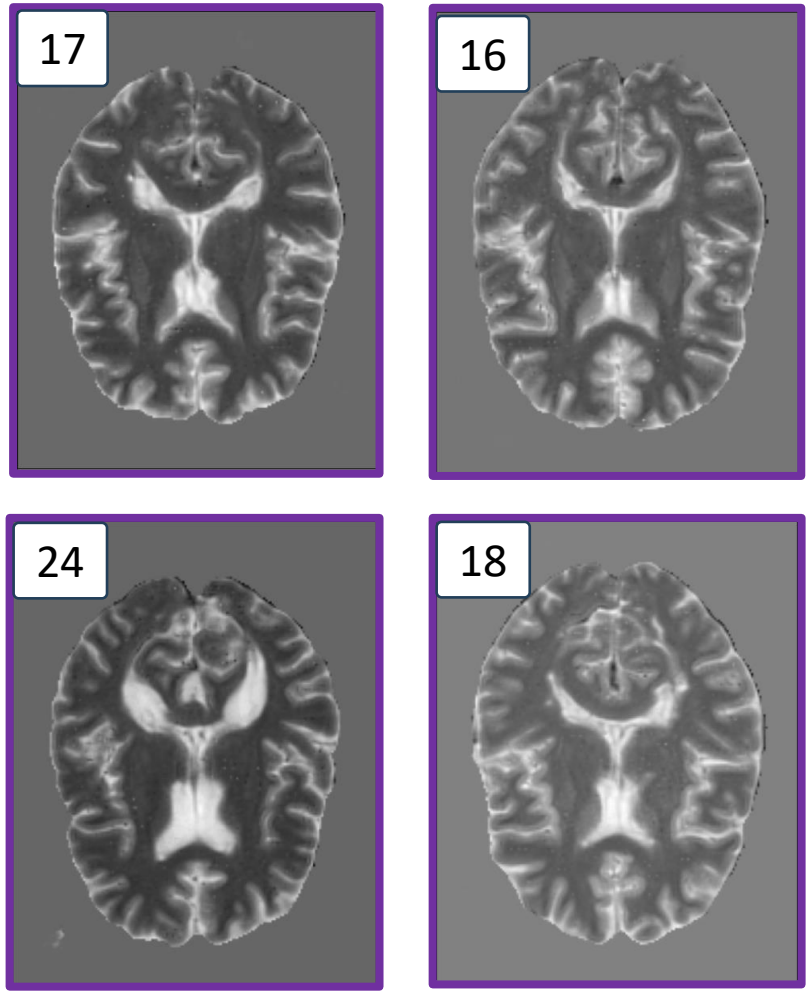
Real



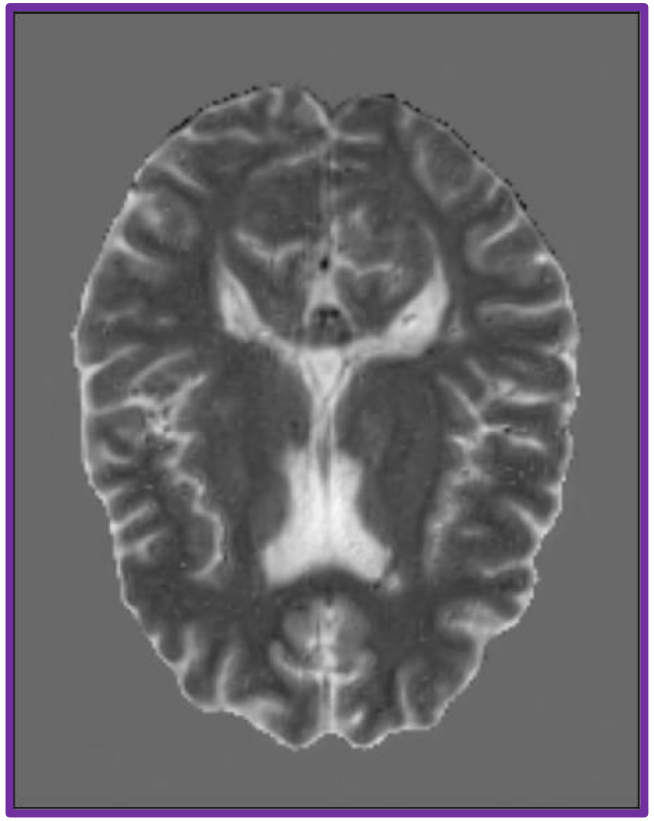
Generated

LDM: Uniqueness

- Visually appear to be unique
- Quantitative metrics:
  - SSIM within synthetic
  - SSIM within ground truth

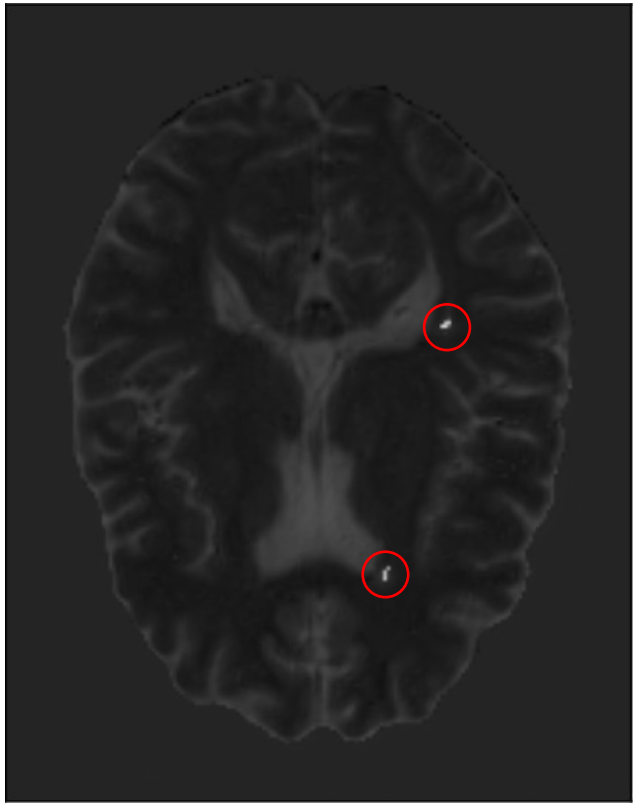


LDM: Pathology



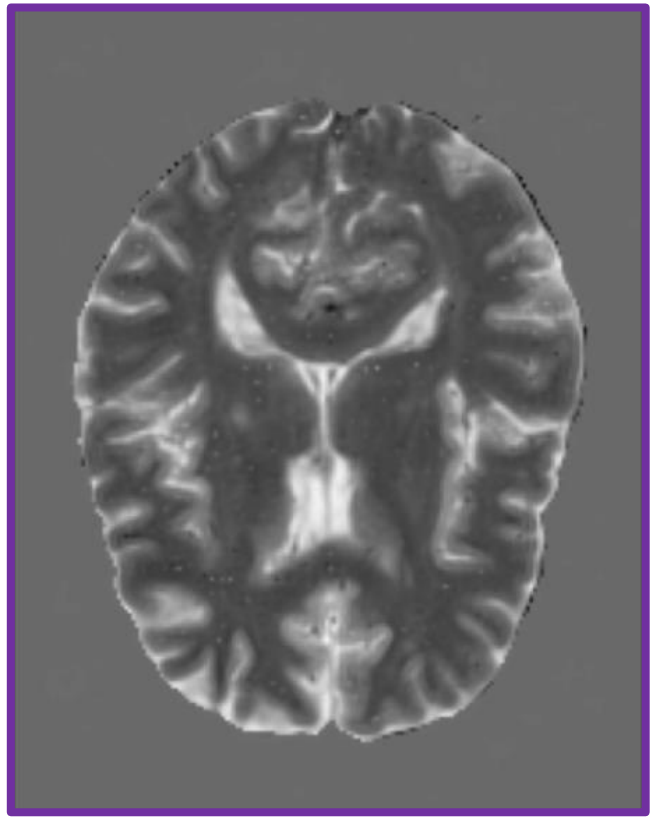
T2-Weighted

Lesion Map

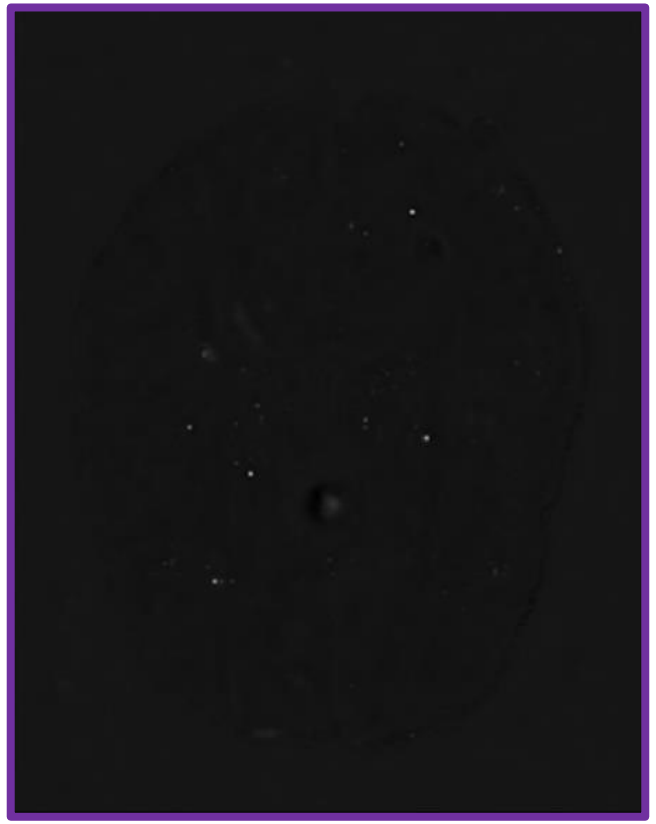


Overlay

LDM: Pathology



T2-Weighted



Lesion Map

- In many cases sequences not aligned
- Future research:
  - Conditioning
- Proxy metric:
  - Pre-trained segmentation model

Conclusions

McGill

Conclusions

- A shortage of MRI data in the MS domain represents a hurdle to research in image-based medicine
- Synthetic image generation presents an avenue around the barriers caused by a lack of data
- Latent diffusion models show capability, but promise is tied to discovering realizable and meaningful evaluation metrics

References

References

[1] Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., ... & Bengio, Y. (2014). Generative adversarial nets. Advances in neural information processing systems, 27.

[2] Tero Karras, Samuli Laine, Timo Aila; Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2019, pp. 4401-4410

[3] Kingma, D. P., & Welling, M. (2013). Auto-encoding variational bayes. arXiv preprint arXiv:1312.6114.

[4] Oord, A. V. D., Vinyals, O., & Kavukcuoglu, K. (2017). Neural discrete representation learning. arXiv preprint arXiv:1711.00937.

[5] Yang, L., Zhang, Z., Song, Y., Hong, S., Xu, R., Zhao, Y., ... & Yang, M. H. (2022). Diffusion models: A comprehensive survey of methods and applications. ACM Computing Surveys.

References

[6] Song, J., Meng, C., & Ermon, S. (2020). Denoising diffusion implicit models. arXiv preprint arXiv:2010.02502.

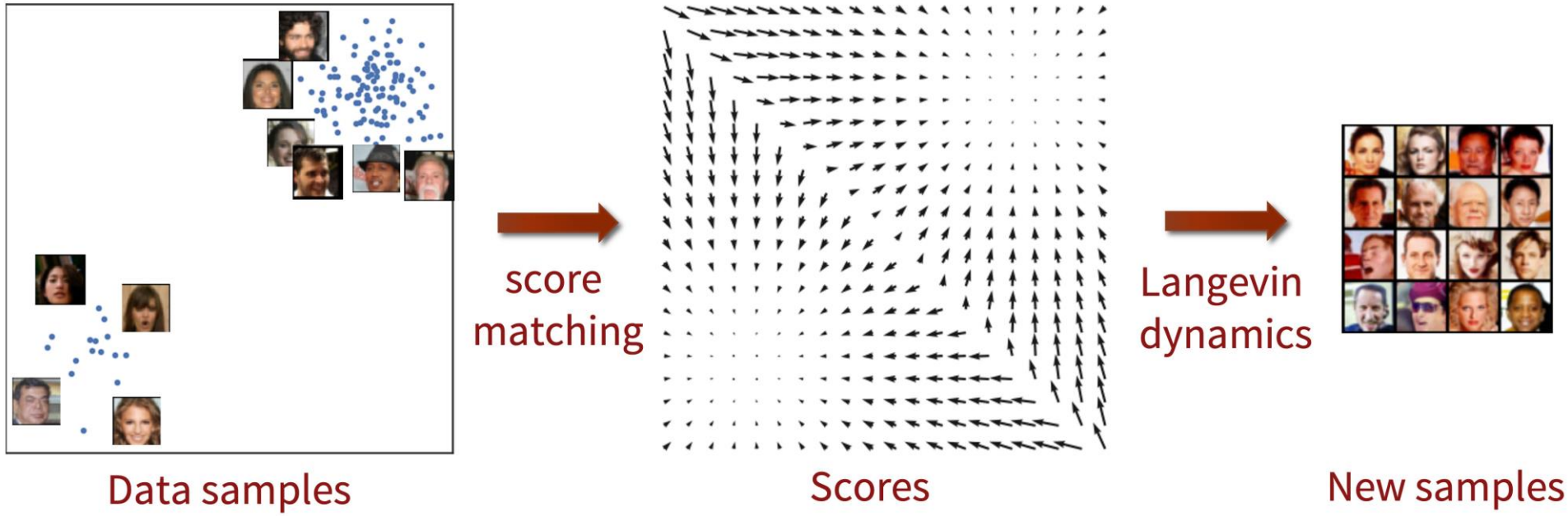
[7] Rombach, R., Blattmann, A., Lorenz, D., Esser, P., & Ommer, B. (2022). High-resolution image synthesis with latent diffusion models. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 10684-10695).

[8] Zhou Wang, A. C. Bovik, H. R. Sheikh and E. P. Simoncelli, "Image quality assessment: from error visibility to structural similarity," in IEEE Transactions on Image Processing, vol. 13, no. 4, pp. 600-612, April 2004, doi: 10.1109/TIP.2003.819861.

Appendix

McGill

Denoising Score Matching

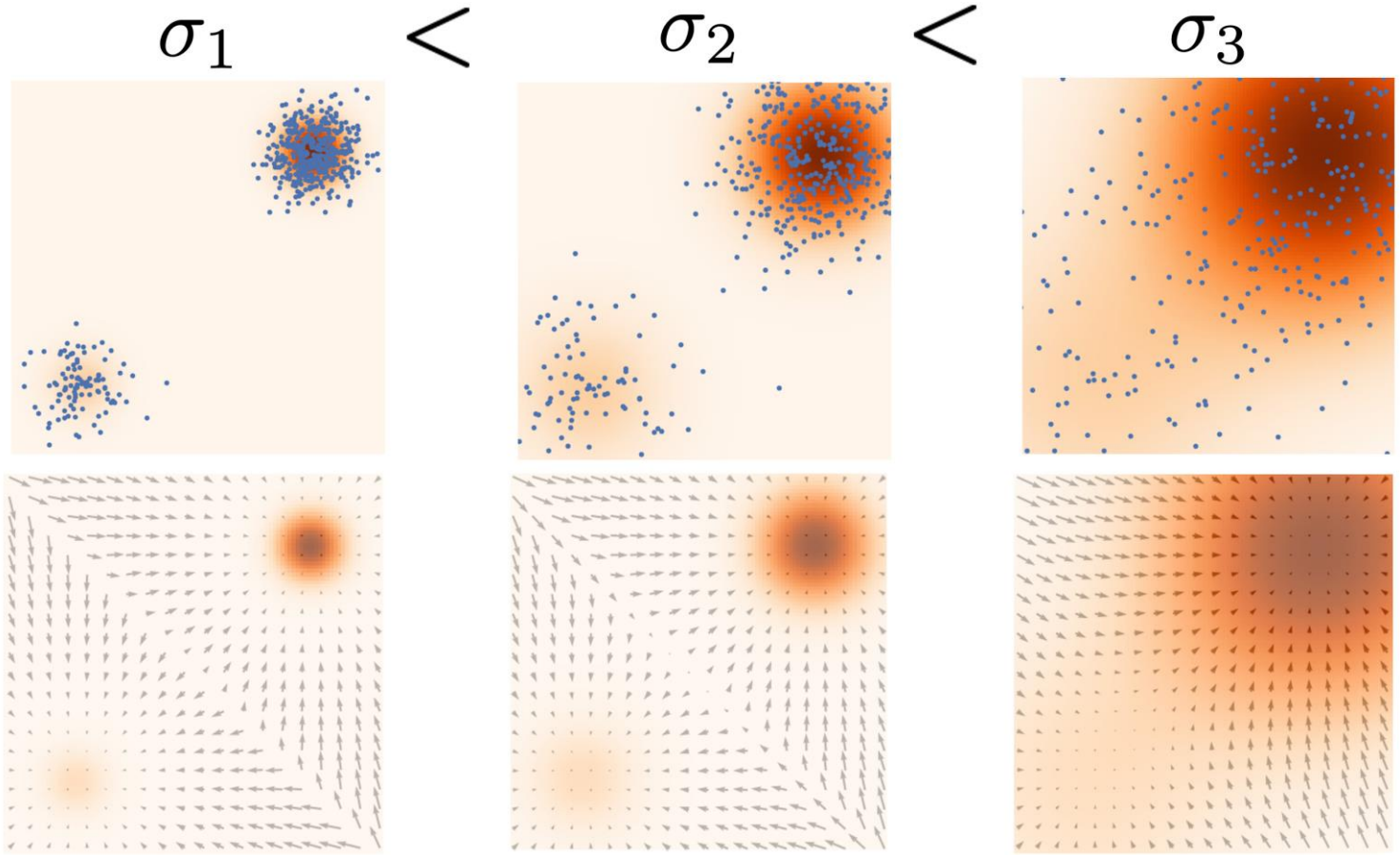


$$\{\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_N\} \overset{\text{i.i.d.}}{\sim} p(\mathbf{x})$$

$$\mathbf{s}_\theta(\mathbf{x}) \approx \nabla_{\mathbf{x}} \log p(\mathbf{x})$$

https://yang-song.net/blog/2021/score/

Denoising Score Matching



https://yang-song.net/blog/2021/score/

Autoencoder: Hyperparameters

Table 2: Autoencoder Training Hyperparameters

Model Component	Parameter	Value
AutoencoderKL	spatial_dims	2
	in_channels	1
	out_channels	1
	num_channels	[128, 128, 256]
	latent_channels	3
	num_res_blocks	2
	attention_levels	[False, False, False]
	with_encoder_nonlocal_attn	False
	with_decoder_nonlocal_attn	False
PatchDiscriminator	spatial_dims	2
	num_layers_d	3
	num_channels	64
	in_channels	2
	out_channels	2
PerceptualLoss	spatial_dims	2
	network_type	'alex'
PatchAdversarialLoss	criterion	'least_squares'
Learning Rate		1e-4